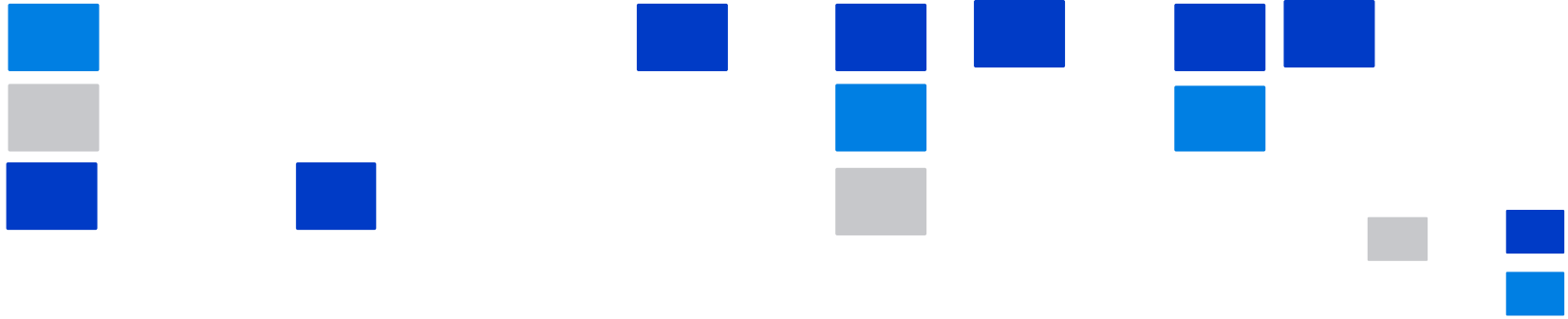




Sistemes Operatius

Tema 4: Crides al sistema - Memòria compartida • **Curs 2026-27**

Memòria compartida

Passos:

1. Crear una regió (`shmget()`).
2. Lligar les adreces (`shmat()`).
3. Usar la memòria compartida.
4. Deslligar les adreces (`shmdt()`).
5. Eliminar la regió (`shmctl()`).

Include utilitzat: `<sys/shm.h>`

Comanda shell interessant: `ipcs`

Exemple Escriptor.c

```
int main() {
    int memid, *num, n;
    char buf[64];

    memid = shmget(0xCAFECAFE, sizeof(int), IPC_CREAT | IPC_EXCL | 0600);
    if (memid != -1) {
        num = shmat(memid, NULL, 0);
        *num = 0;
        while (*num != -1) {
            write(STDOUT_FILENO, "Number: ", 8);
            n = read(STDIN_FILENO, buf, sizeof(buf) - 1);
            if (n <= 0) break;
            buf[n] = '\\0';
            *num = (int)strtol(buf, NULL, 10);
        }
        shmdt(num);
        shmctl(memid, IPC_RMID, NULL);
    } else {
        write(STDOUT_FILENO, "Existing region\\n", 16);
    }
    return 0;
}
```

Exemple Lector.c

```
int main() {
    int memid, *num, n;
    char buf[64];
    char c;

    memid = shmget(0xCAFECAFE, 0, 0);
    if (memid != -1) {
        num = shmat(memid, NULL, 0);
        while (*num != -1) {
            n = sprintf(buf, "Number: %d\n", *num);
            write(STDOUT_FILENO, buf, n);

            read(STDIN_FILENO, &c, 1);
        }
        shmdt(num);
    } else {
        write(STDOUT_FILENO, "Region does not exist\n", 22);
    }
    return 0;
}
```